# Time series: properties of ARMA processes

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The homework is provided on the final slides

# Causal AR(1) process

An AR(1) process $x_t$ satisfies the equation

$$x_t = \alpha x_{t-1} + \epsilon_t$$

for all $t$. Using this equation with $t \Leftarrow t-1$, we have $x_{t-1} = \alpha x_{t-2} + \epsilon_{t-1}$ and then we obtain

$$x_t = \alpha(\alpha x_{t-2} + \epsilon_{t-1}) + \epsilon_t = \alpha^2 x_{t-2} + \alpha\epsilon_{t-1} + \epsilon_t.$$

Proceeding further, successive substitutions yield

$$\begin{aligned} x_t &= \alpha^3 x_{t-3} + \alpha^2\epsilon_{t-2} + \alpha\epsilon_{t-1} + \epsilon_t \\ &= \alpha^4 x_{t-4} + \alpha^2\epsilon_{t-3} + \alpha^2\epsilon_{t-2} + \alpha\epsilon_{t-1} + \epsilon_t \\ &= \sum_{j=0}^{\infty} \alpha^j \epsilon_{t-j} \quad \text{(converges if } |\alpha| < 1\text{)}. \end{aligned}$$

Thus, an AR(1) process can be written as an MA($\infty$) process. Condition $|\alpha| < 1$ means "no roots" of $a(z) = 1 - \alpha z$ within the unit circle, see slide 7 of Lecture notes 8.

# Causal AR(1) process as MA: another way

Using the backward shift operator B, an AR(1) process can be written as

$$(1 - \alpha B)x_t = \epsilon_t.$$

Since $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$, where $|u| < 1$, we get

$$\begin{aligned} x_t &= \frac{1}{1 - \alpha B}\epsilon_t = (1 + \alpha B + \alpha^2 B^2 + \ldots)\epsilon_t \\ &= \epsilon_t + \alpha\epsilon_{t-1} + \alpha^2\epsilon_{t-2} + \ldots. \end{aligned}$$

Theorem. An AR(1) process $x_t$ satisfying $x_t = \alpha x_{t-1} + \epsilon_t$ has the following equivalent representation

$$x_t = \sum_{j=0}^{\infty} \alpha^j \epsilon_{t-j}.$$

# Causal AR($p$) process as MA

Write an AR($p$) process as $a(B)x_t = \epsilon_t$, that is,

$$(1 - a_1 B - \ldots - a_p B^p)x_t = \epsilon_t.$$

Using the expansion $\frac{1}{1-u} = 1 + u + u^2 + u^3 + \ldots$ with $u = a_1 B + \ldots + a_p B^p$, we obtain

$$\begin{aligned} x_t = \epsilon_t &+ (a_1 B + \ldots + a_p B^p)\epsilon_t \\ &+ (a_1 B + \ldots + a_p B^p)^2 \epsilon_t \\ &+ (a_1 B + \ldots + a_p B^p)^3 \epsilon_t + \ldots \end{aligned}$$

Thus

$$\begin{aligned} x_t = \epsilon_t &+ a_1 \epsilon_{t-1} + \ldots + a_p \epsilon_{t-p} \\ &+ \sum_{i=1}^{p} \sum_{j=1}^{p} a_i a_j \epsilon_{t-i-j} \\ &+ \sum_{i=1}^{p} \sum_{j=1}^{p} \sum_{s=1}^{p} a_i a_j a_s \epsilon_{t-i-j-s} + \ldots \end{aligned}$$

# Covariance function of AR(1) process

Consider an AR(1) process $x_t$ satisfying $x_t = \alpha x_{t-1} + \epsilon_t$. Using the MA($\infty$) representation and since $\mathbf{E}\epsilon_t\epsilon_s = \begin{cases} \sigma^2, & t=s \\ 0, & t \neq s \end{cases}$, we have

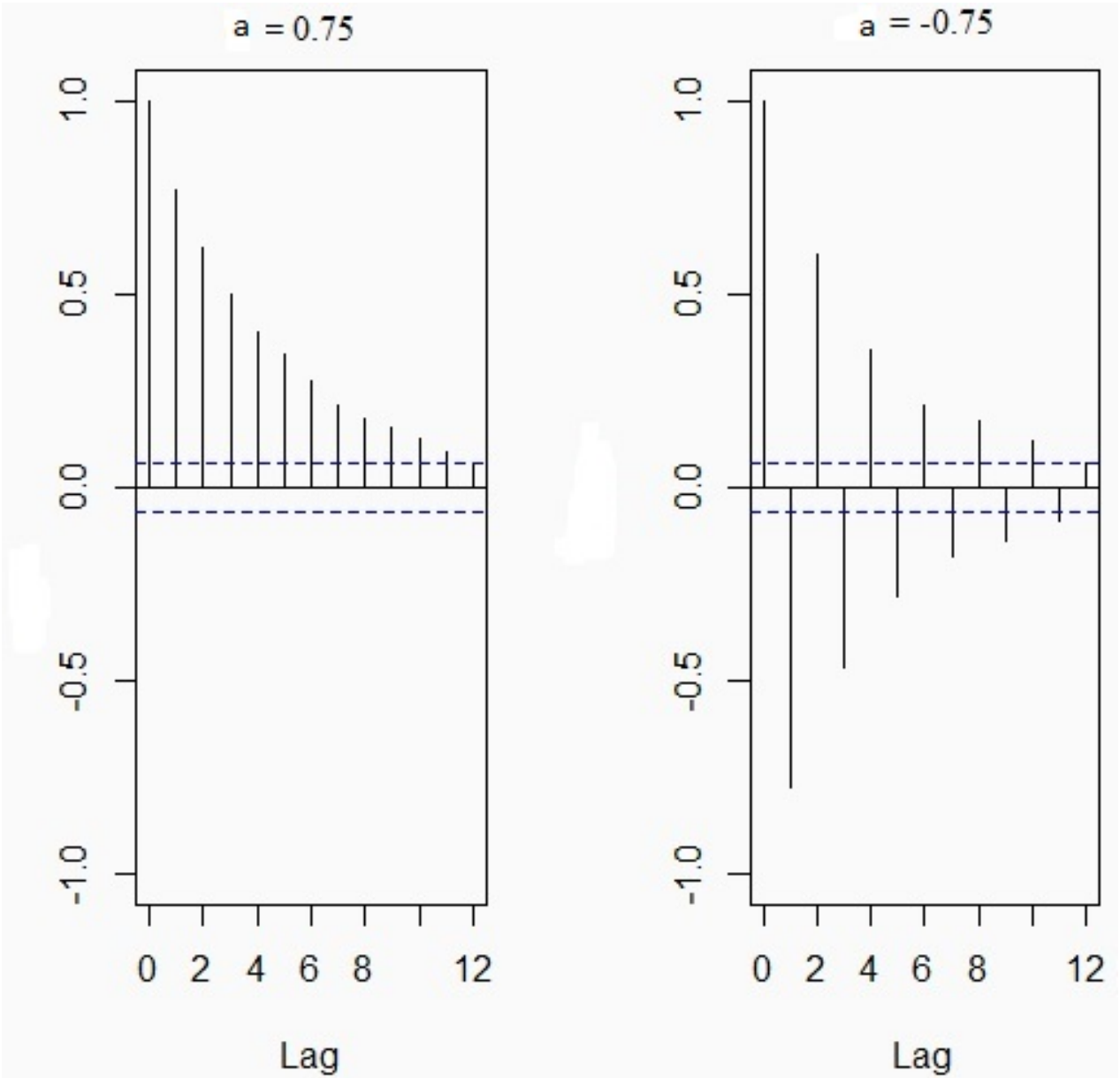
$$R(k) = \mathbf{E}x_t x_{t+k} = \mathbf{E}\Big(\sum_{j=0}^{\infty} \alpha^j \epsilon_{t-j}\Big)\Big(\sum_{i=0}^{\infty} \alpha^i \epsilon_{t+k-i}\Big)$$

$$= \sum_{j=0}^{\infty}\sum_{i=0}^{\infty} \alpha^j \alpha^i \mathbf{E}\epsilon_{t-j}\epsilon_{t+k-i} = \sigma^2 \sum_{j=0}^{\infty} \alpha^j \alpha^{k+j}$$

$$= \sigma^2 \alpha^k \sum_{j=0}^{\infty} \alpha^{2j} = \sigma^2 \alpha^k \frac{1}{1-\alpha^2}.$$

Theorem. The variance of the AR(1) process is $R(0) = \frac{\sigma^2}{1-\alpha^2}$ and the correlation function is $\rho(k) = \alpha^k$, $k = 0, 1, 2, \ldots$.

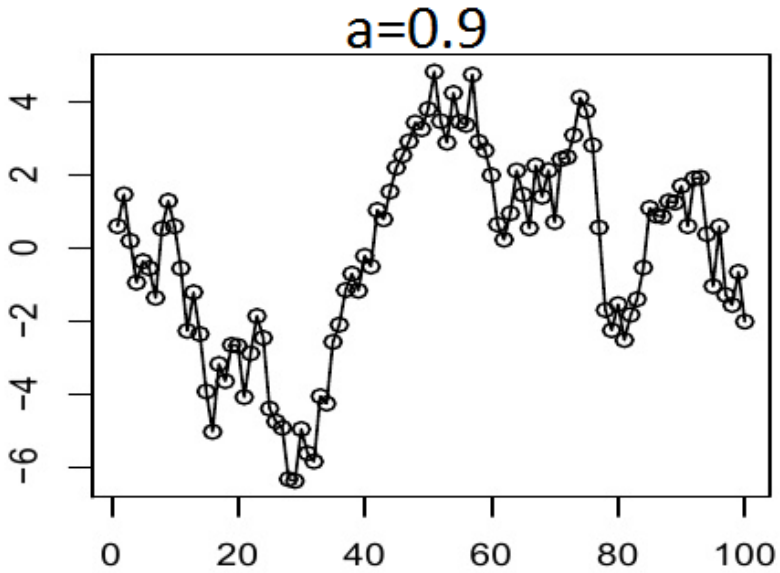
# Correlogram of AR(1) process

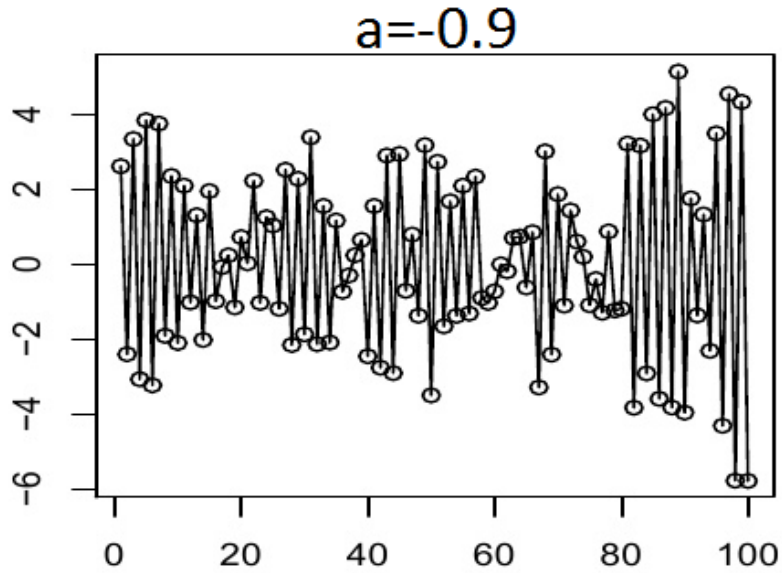


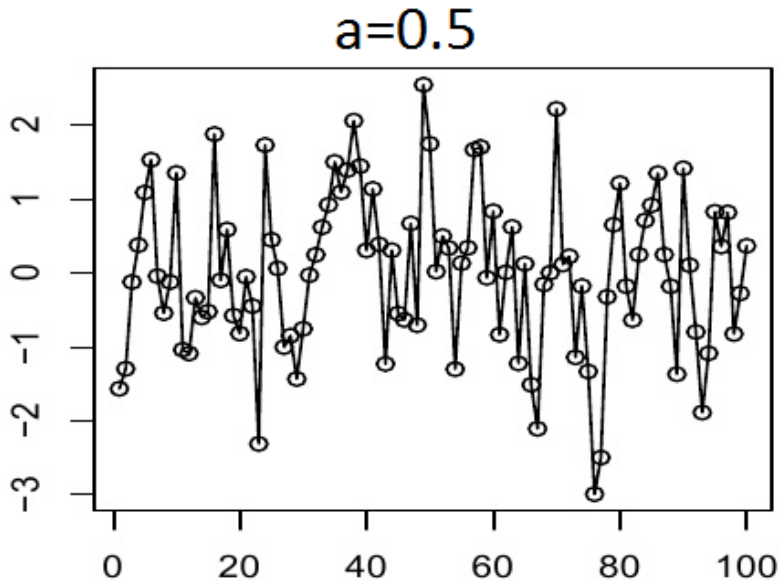
# Realizations of AR(1) process

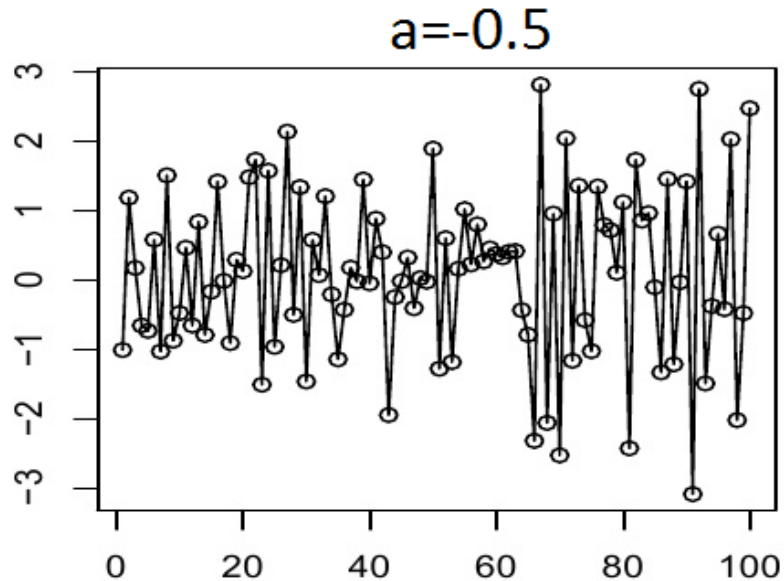
```
® SimulAR=function(N,p,a,initialV,Sigma)
{ eps=rnorm(N,0,Sigma); x=rep(0,N);
for(t in c(1:p)) { x[t]=initialV[t]; }
for(t in c((p+1):N)) { x[t]=sum(-a[2:(p+1)]*x[t-c(1:p)])+eps[t];}
return(x); }
n=100; sigma=1.2; par(mfrow=c(2,2));
x=SimulAR(n,1,c(1,-0.8),c(1.1),sigma);
plot((1:n), x, pch=19, main="AR(1) process a=(1,-0.8) a(z)=1-0.8z");
lines((1:n),x);
x=SimulAR(n,1,c(1,-0.4),c(1.1),sigma);
plot((1:n), x, pch=19, main="AR(1) process a=(1,-0.4) a(z)=1-0.4z");
lines((1:n),x);
x=SimulAR(n,1,c(1,0.3),c(1.1),sigma);
plot((1:n), x, pch=19, main="AR(1) process a=(1,0.3) a(z)=1+0.3z");
lines((1:n),x);
x=SimulAR(n,1,c(1,0.7),c(1.1),sigma);
plot((1:n), x, pch=19, main="AR(1) process a=(1,0.7) a(z)=1+0.7z");
lines((1:n),x);
```

# Realizations of AR(1) process



# Causality of AR(2) process

An AR(2) process given by

$$x_t = a_1 x_{t-1} + a_2 x_{t-2} + \epsilon_t.$$

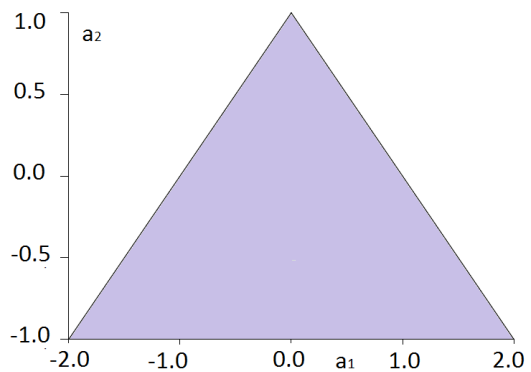
is causal if the polynomial $a(z) = 1 - a_1 z - a_2 z^2$ has no roots within the unit circle. Roots of $a(z)$ are given by

$$z_1 = \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{-2a_2}, \quad z_2 = \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{-2a_2}.$$

**Theorem**

The AR(2) process is causal (i.e. $|z_1| > 1$ and $|z_2| > 1$) **iff**

$$a_2 + a_1 < 1, \quad a_2 - a_1 < 1, \quad |a_2| < 1.$$



# Inequalities for coefficients of causal AR(2)

$a(z) = 1 - a_1 z - a_2 z^2$ has no roots within the unit circle **iff** $\tilde{a}(u) = u^2 - a_1 u - a_2$ has no roots outside the unit circle.
Roots of $\tilde{a}(u)$ are given by

$$u_1 = \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2}, \quad u_2 = \frac{a_1 - \sqrt{a_1^2 + 4a_2}}{2}.$$

Condition $|u_1| < 1$ can be rewritten as $-1 < \frac{a_1 + \sqrt{a_1^2 + 4a_2}}{2} < 1$. Then $-2 < a_1 + \sqrt{a_1^2 + 4a_2} < 2$. Thus, $-2 - a_1 < \sqrt{a_1^2 + 4a_2} < 2 - a_1$.

If $a_1^2 + 4a_2 > 0$, then $2 - a_1 > 0$, $a_1^2 + 4a_2 < (2 - a_1)^2$,
that is, $2 > a_1$, $a_2 + a_1 < 1$.

If $a_1^2 + 4a_2 < 0$, then $u_1$ is a complex number and $|u_1|^2 < 1$. Then $|a_1 + i\sqrt{|a_1^2 + 4a_2|}|^2 < 2^2$, then $a_1^2 - (a_1^2 + 4a_2) < 4$, thus $a_2 > -1$.

Similarly, condition $|u_2| < 1$ can be considered.

# Correlation function of AR(2): via roots

For the AR(2) process $x_t$ satisfying $x_t = a_1 x_{t-1} + a_2 x_{t-2} + \epsilon_t$, the $k$-th Yule-Walker equation for the correlation function is

$$\rho(k) = a_1 \rho(k-1) + a_2 \rho(k-2), \quad k = 1, 2, 3, \ldots.$$

From theory of linear recurrent equations, it follows that

Theorem. The correlation function $\rho(k)$ of AR(2) process is

$$\rho(k) = \begin{cases} A_1 z_1^{|k|} + A_2 z_2^{|k|} & z_1, z_2 \text{ are real and distinct} \\ A_1 (1 + A_2 k) z_1^{|k|} & z_1, z_2 \text{ are real and } z_1 = z_2 \\ A_1 z_{1re}^{|k|} \cos(z_{1im} k + A_2) & z_1, z_2 \text{ are complex} \end{cases}$$

for all $k$, where $z_1, z_2$ are roots of $a(1/z) = 1 - a_1 z^{-1} - a_2 z^{-2}$ and $A_1, A_2$ are some constants, $z_1 = z_{1re} + i z_{1im}$.

Constants $A_1, A_2$ can be found from the 1-st Yule-Walker equation.

# Correlation function of AR(2): constants

The 1-st Yule-Walker equation is

$$\rho(1) = a_1\rho(0) + a_2\rho(-1).$$

Since $\rho(0) = 1$ and $\rho(-1) = \rho(1)$, we have

$$\rho(1) = a_1 + a_2\rho(1) \quad \text{or} \quad \rho(1) = \frac{a_1}{1 - a_2}.$$

Consider the case when $z_1, z_2$ are real and distinct.
From $\rho(0) = A_1 z_1^0 + A_2 z_2^0 = 1$, we have $A_2 = 1 - A_1$. Then

$$\frac{a_1}{1 - a_2} = \rho(1) = A_1 z_1^1 + A_2 z_2^1 = A_1 z_1 + (1 - A_1) z_2.$$

Then $\frac{a_1}{1-a_2} - z_2 = A_1(z_1 - z_2)$, and $A_1 = [\frac{a_1}{1-a_2} - z_2]/(z_1 - z_2)$.

# Correlation function of AR(2): via iterations

Another way to compute $\rho(k)$ for an AR(2) process with given coefficients $a_1, a_2$ is based on the following sequential iterations.
From the system of the 1-st and 2-nd Yule-Walker equations

$$\begin{cases} \rho(1) = a_1 + a_2\rho(1) \\ \rho(2) = a_1\rho(1) + a_2 \end{cases}$$

**Theorem**

For AR(2) we have $\rho(1) = \dfrac{a_1}{1-a_2}$ and $\rho(2) = \dfrac{a_1^2}{1-a_2} + a_2$. Also from the $k$-th Yule-Walker equations, we have

$$\rho(k) = a_1\rho(k-1) + a_2\rho(k-2), \quad k = 3, 4, 5, \ldots$$

where $\rho(k-1)$ and $\rho(k-2)$ are computed in previous iterations.

# Correlation function of AR($p$): via roots

For an AR($p$) process $x_t$ satisfying

$$x_t = a_1 x_{t-1} + a_2 x_{t-2} + \ldots + a_p x_{t-p} + \epsilon_t$$

from theory of linear recurrent equations, it follows that $\rho(k)$ has the form

$$\rho(k) = A_1 z_1^{|k|} + A_2 z_2^{|k|} + \ldots + A_p z_p^{|k|}$$

for all $k$, where $z_1, z_2, \ldots, z_p$ are roots of

$$a(1/z) = 1 - a_1 z^{-1} - \ldots - a_p z^{-p}$$

and $A_1, A_2, \ldots, A_p$ are some constants, if $z_i \neq z_j$ for all $i, j$.

# Correlation function of AR($p$): via iterations

Computing $\rho(k)$ for an AR($p$) process with given values $a_1, a_2, \ldots, a_p$ can be done as follows.
From the system of the $1, 2, \ldots, p$-th Yule-Walker equations

$$\begin{cases} \rho(1) = & a_1 & + & a_2\rho(1) & + \ldots + & a_p\rho(p-1) \\ \rho(2) = & a_1\rho(1) & + & a_2 & + \ldots + & a_p\rho(p-2) \\ \rho(3) = & a_1\rho(2) & + & a_2\rho(1) & + \ldots + & a_p\rho(p-3) \\ \vdots & & & & & \\ \rho(p) = & a_1\rho(p-1) & + & a_2\rho(p-2) & + \ldots + & a_p \end{cases}$$

we get $\rho(1), \ldots, \rho(p)$ as expressions of $a_1, \ldots, a_p$.
Then from the $k$-th Yule-Walker equations, we get

$$\rho(k) = a_1\rho(k-1) + a_2\rho(k-2) + \ldots + a_p\rho(k-p), \quad k = p+1, p+2, \ldots$$

where $\rho(k-1), \ldots, \rho(k-p)$ are computed in previous iterations.

# Simulation of AR(2) process

Using the algorithm from slide 3 of Lecture 8, simulate realizations of length $n = 100$ from an ARMA(2,0) process with

- $a(z) = 1 - 0.5z + 0.5z^2$, complex roots
- $a(z) = 1 - 1.1z + 0.3z^2$, two distinct real roots
- $a(z) = 1 + 0.5z - 0.25z^2$, two distinct real roots
- $a(z) = 1 - z + 0.5z^2$, complex roots

and the initial conditions $x_1 = 1.1$ and $x_2 = 0.4$, $\epsilon_t$ is Gaussian white noise with mean 0 and $\sigma^2 = 1.2^2$.

```
® acoef=c(1,-0.5,0.5);
aroots=polyroot(acoef);
print("polynomial with coefficients"); print(acoef);
print("has roots"); print(aroots);
```

# Simulation of AR(2) process

```
® SimulAR=function(N,p,a,initialV,Sigma)
{ eps=rnorm(N,0,Sigma); x=rep(0,N);
for(t in c(1:p)) { x[t]=initialV[t]; }
for(t in c((p+1):N)) { x[t]=sum(-a[2:(p+1)]*x[t-c(1:p)])+eps[t];}
return(x); }
n=100; sigma=1.2; par(mfrow=c(2,2));
x=SimulAR(n,2,c(1,-0.5,0.5),c(1.1,0.4),sigma);
plot((1:n), x, pch=19, main="AR(2) process a=(1,-0.5,0.5)");
lines((1:n),x);
x=SimulAR(n,2,c(1,-1.1,0.3),c(1.1,0.4),sigma);
plot((1:n), x, pch=19, main="AR(2) process a=(1,-1.1,0.3)");
lines((1:n),x);
x=SimulAR(n,2,c(1,0.5,-0.25),c(1.1,0.4),sigma);
plot((1:n), x, pch=19, main="AR(2) process a=(1,0.5,-0.25)");
lines((1:n),x);
x=SimulAR(n,2,c(1,-1,0.5),c(1.1,0.4),sigma);
plot((1:n), x, pch=19, main="AR(2) process a=(1,-1,0.5)");
lines((1:n),x);
```

# Homework: Learning with AI

Utilize AI tools to explore the deeper properties of AR processes.

- **MA($\infty$) Representation:** Ask an AI to perform the successive back-substitution for an AR(2) process (e.g., $x_t = 0.5x_{t-1} + 0.1x_{t-2} + \epsilon_t$) to find the first four terms of its MA representation. Compare the effort with the operator method $(1 - a_1B - a_2B^2)^{-1}$.
- **Visualizing ACF Behavior:** Ask the AI to generate R-code that plots the ACF for two different AR(2) models: one with real roots and one with complex roots. Observe and explain the difference between exponential decay and sinusoidal oscillations (pseudo-periods).
- **Code Optimization:** Take the `SimulAR` function from Slide 17 and ask the AI to "vectorize" the loop or use a built-in R function like `arima.sim` to achieve the same result. Compare the outputs to ensure consistency.

# Monday's challenging homework

- Use AI to learn how to record video using Zoom.
- Use AI to learn how to create a YouTube channel and upload videos.
- Create a secondary Google account and a corresponding YouTube channel without using your surname and student number.
- Record and upload a video demonstrating the use of the AI prompt that generates an R script for fitting an ARMA model to annual earthquake data and executing the script in RStudio.
- Write the link to your video in the 'Discussions' tab on Learning Central.

# Homework (part 2)

- Prepare a collection of graphs with realizations of AR(1) and AR(2) processes with different parameters.
- For each realization, depict the correlogram and present values $\hat{\rho}(1), \hat{\rho}(2), \hat{\rho}(3)$.
- Solve exam-like exercises for L9.